

# CYBERSECURITY: LABS

## LAB 2B: GNUPG WITH PKI

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# OVERVIEW

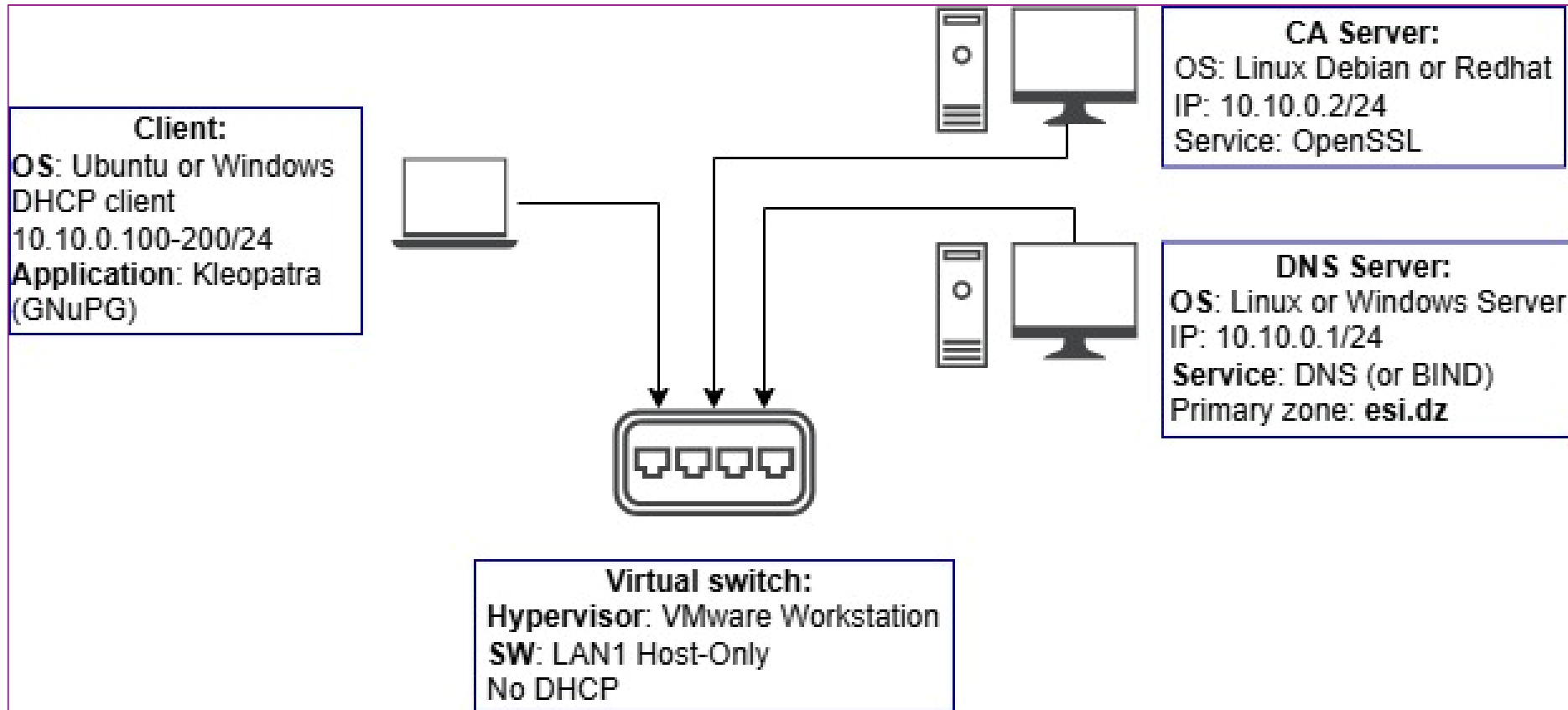
1. Objectives
2. Lab Environnement
3. Preparation
4. Creating S/MIME certificate request on Kleopatra
5. Signing the S/MIME request by the CA
6. Installing the S/MIME certicate on Kleopatra
7. Checking installation
8. For students

# OBJECTIVES

- The main objective of this Lab is to use a **CA-signed user certificate**, instead of using a **self-signed certificate** (default configuration) in **Kleopatra tool**.
- For **N** users, we need:

| Certificate type | Public keys Number | Total                             | Example :<br><b>N=100</b>    |
|------------------|--------------------|-----------------------------------|------------------------------|
| Self-signed      | <b>N</b> per user  | <b>N<sup>2</sup></b> in all users | <b>10<sup>4</sup>=10,000</b> |
| CA-signed        | <b>1</b> per user  | <b>N</b> in the CA                | <b>100</b>                   |

# VM ENVIRONMENT



# PREPARATION

- Using the **ESI CA** (created in Lab2A with OpenSSL) on [**CA Server VM**]
- Install GnuPG with **Kleopatra** on [**Win10\_Client VM**]
- Have **DNS service** available on [**CA Server VM**]

# CREATING AN S/MIME CERTIFICATE FOR A USER

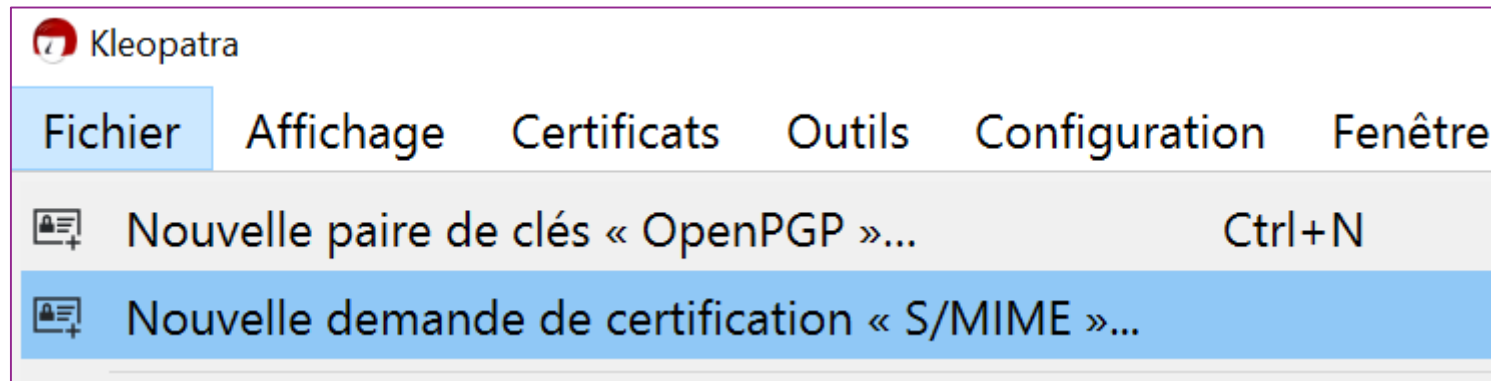
- An **S/MIME** (Secure/Multipurpose Internet Mail Extensions) certificate is a **user digital certificate** used to encrypt and sign emails, ensuring data privacy, message integrity, and sender authentication.
- **Note that we have 3 main types of certificate:**
  - **Computer** certificate (server/machine: SSL/TLS certificate)
  - **User** certificate (client: S/MIME certificate)
  - **Service** certificate (Code-signing)

# USER CERTIFICATE CREATION PROCESS

- **Method 1:** from kleopatra [Win10\_Client VM]
- **Method 2:** from OpenSSL [Win10\_Client VM]
  - Install **XAMPP** (it installs also OpenSSL)
  - Install OpenSSL source from Github:  
<https://github.com/openssl/openssl>

# METHOD 1: S/MIME CERTIFICATE FROM KLEOPATRA

- In **Kleopatra**, select **New S/MIME certification request**:



# FILL THE FORM

?

×

Assistant de création de paire de clés

## Saisir les renseignements

Veuillez saisir vos renseignements personnels ci-dessous. Si vous désirez contrôler davantage de paramètres, cliquez sur le bouton « Configuration avancée ».

|                                            |                                             |              |
|--------------------------------------------|---------------------------------------------|--------------|
| Nom complet (CN) :                         | <input type="text" value="Eden"/>           | (requis)     |
| Adresse de courrier électronique (EMAIL) : | <input type="text" value="eden@mylab.net"/> | (requis)     |
| Emplacement (L) :                          | <input type="text" value="ALGER"/>          | (facultatif) |
| Unité d'organisation (OU) :                | <input type="text" value="IT"/>             | (facultatif) |
| Organisation (O) :                         | <input type="text" value="MYLAB"/>          | (facultatif) |
| Indicatif du pays (C) :                    | <input type="text" value="DZ"/>             | (facultatif) |

CN=Eden,L=ALGER,OU=IT,O=MYLAB,C=DZ

Configuration avancée...

Créer

Annuler

# ADVANCED CONFIGURATION

- In the form window, Click on **Advanced Configuration** and choose **2048 bits for RSA**:

Configuration avancée - Kleopatra

Renseignements techniques Renseignements personnels

Clé matérielle

☒ RSA 2 048 bits

☐ + RSA 3 072 bits

☐ DSA 2 048 bits

☐ + Elgamal 2 048 bits

☐ ECDSA/EdDSA ed25519

☐ + ECDH cv25519

Utilisation du certificat

☒ Signature

☒ Chiffrement

OK Annuler

# ADVANCED CONFIGURATION [OPTIONAL]

Configuration avancée - Kleopatra

Renseignements techniques Renseignements personnels

Adresses de courriel

eden@mylab.net

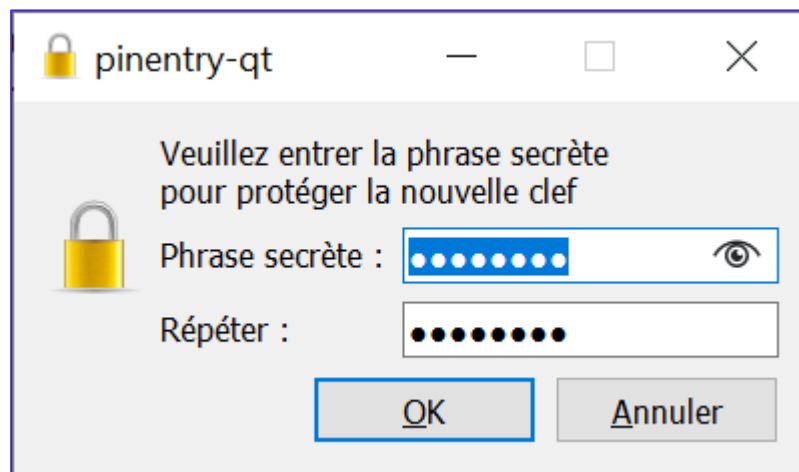
Noms des DNS

mylab.net

URIs

OK Annuler

# PIN CODE FOR THE PRIVATE KEY



A screenshot of a Qt-based pinentry dialog box titled 'pinentry-qt'. The window has a standard title bar with a lock icon, a minus sign, a maximize button, and a close button. The main content area has a light gray background. At the top, it says 'Veuillez entrer la phrase secrète pour protéger la nouvelle clef' (Please enter the secret phrase to protect the new key). Below this, there is a yellow padlock icon. To the right of the padlock, the text 'Phrase secrète :' is followed by a text input field containing ten white dots. To the right of the input field is an eye icon. Below the first input field, the text 'Répéter :' is followed by another text input field containing ten black dots. At the bottom right, there are two buttons: 'OK' and 'Annuler'.

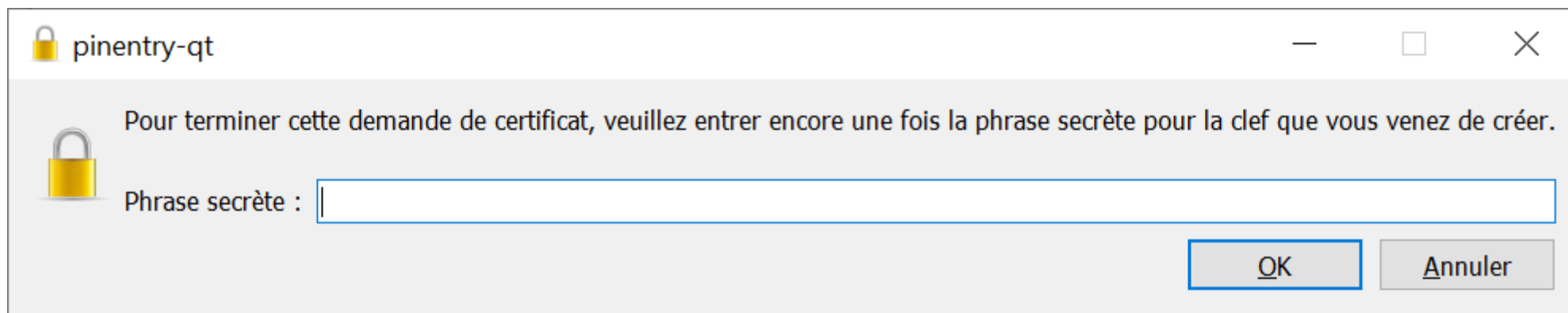
pinentry-qt

Veuillez entrer la phrase secrète pour protéger la nouvelle clef

Phrase secrète : [input field with 10 dots] [eye icon]

Répéter : [input field with 10 dots]

OK Annuler



A screenshot of a Qt-based pinentry dialog box titled 'pinentry-qt'. The window has a standard title bar with a lock icon, a minus sign, a maximize button, and a close button. The main content area has a light gray background. At the top, it says 'Pour terminer cette demande de certificat, veuillez entrer encore une fois la phrase secrète pour la clef que vous venez de créer.' (To finish this certificate request, please enter the secret phrase again for the key you just created). Below this, there is a yellow padlock icon. To the right of the padlock, the text 'Phrase secrète :' is followed by a long, empty text input field. At the bottom right, there are two buttons: 'OK' and 'Annuler'.

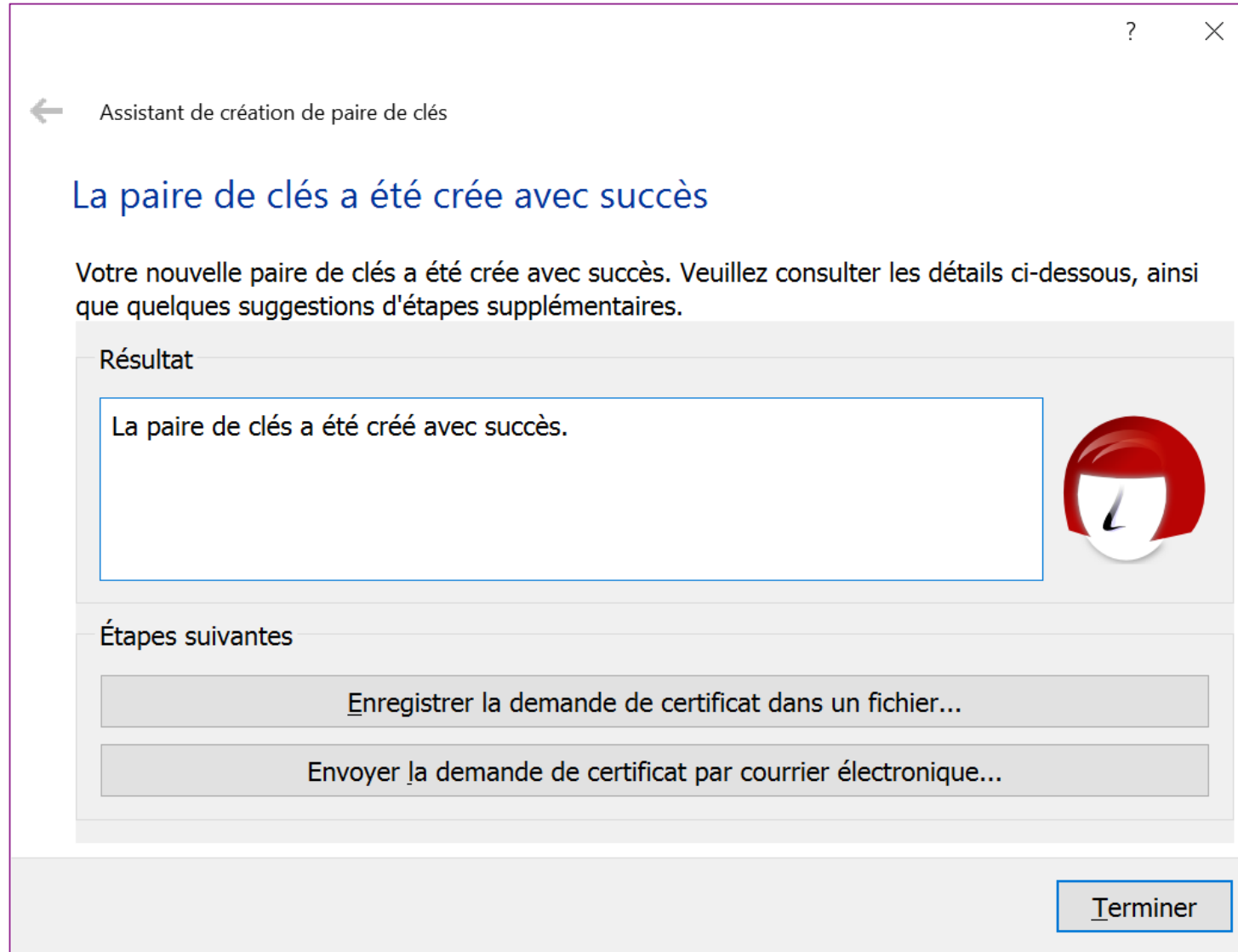
pinentry-qt

Pour terminer cette demande de certificat, veuillez entrer encore une fois la phrase secrète pour la clef que vous venez de créer.

Phrase secrète : [input field]

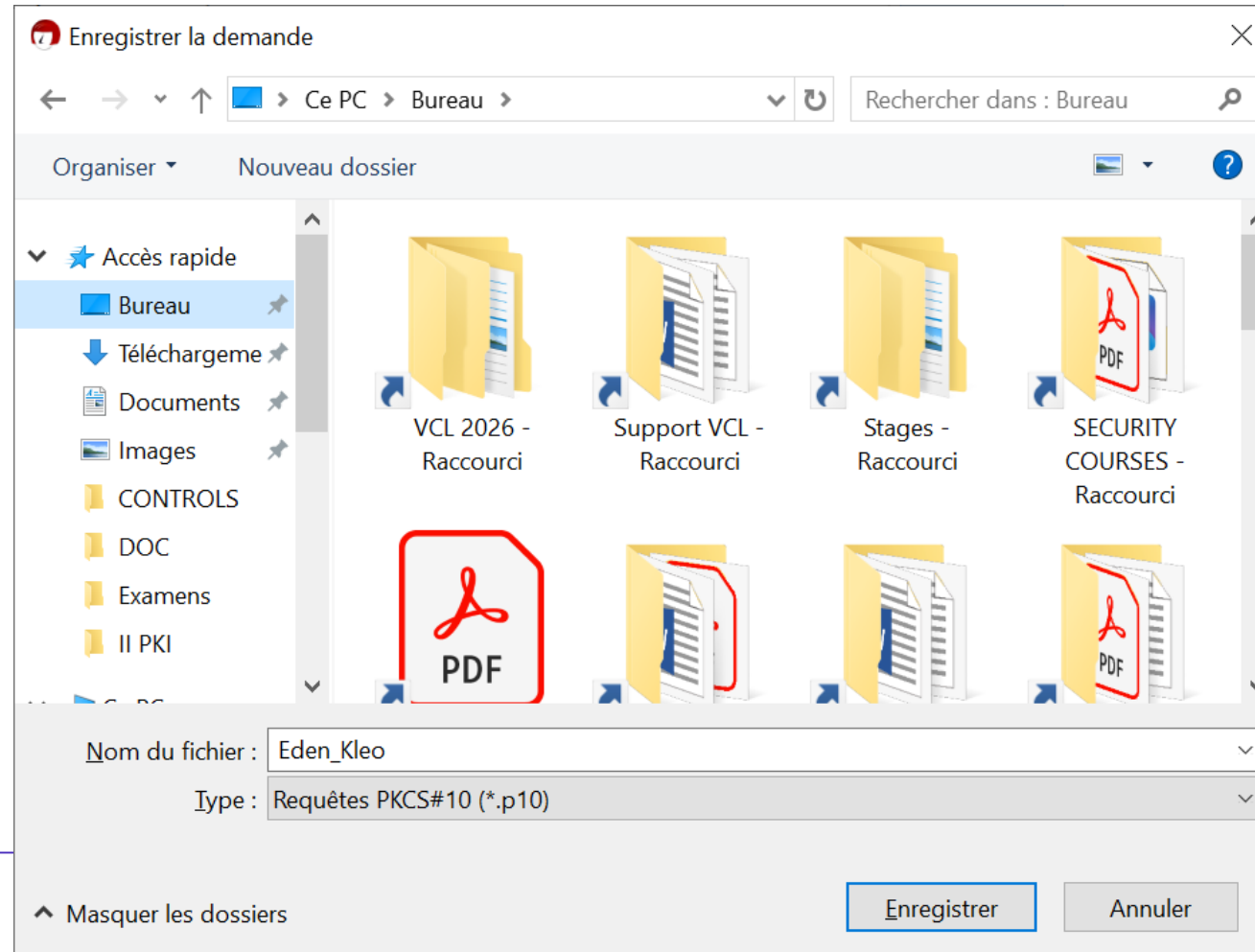
OK Annuler

# THE KEY PAIR IS CREATED



# SAVING THE REQUEST

- Click on *Save the certificate request in a file..* [PKCS#10]



# CREATING USER CERTIFICATE ON THE CA SERVER

**## copy in the correct directory:**

```
cp eden_kleo.p10 /root/ca/certs
```

**## convert to .pem format:**

```
openssl req -inform DER -in certs/eden_kleo.p10 -out certs/eden_kleo.pem -text
```

**## Sign the request by the CA:**

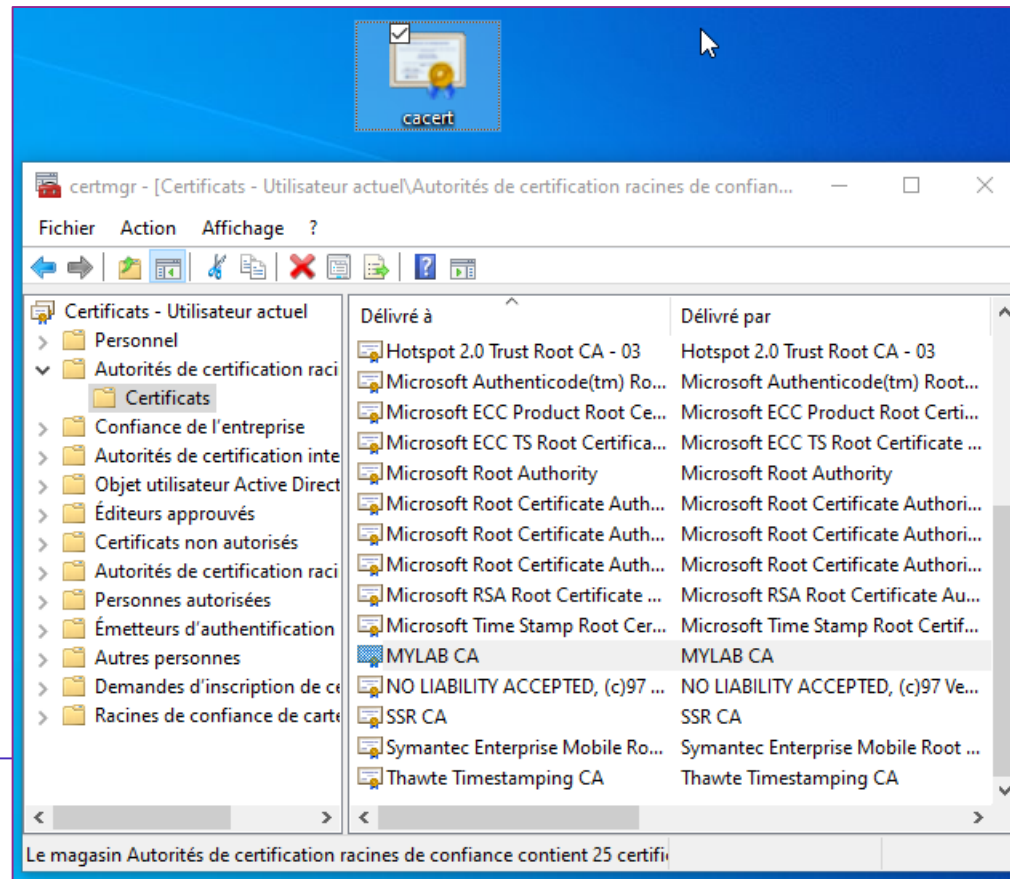
```
openssl ca -config openssl.cnf -policy policy_anything -out certs/eden.pem -infiles  
certs/eden_kleo.pem
```

# COPY THE CA CERTIFICATE TO WINDOWS CERTIFICATES STORE

## On [CA\_Server VM], convert **cacert.pem** from PEM to CER:

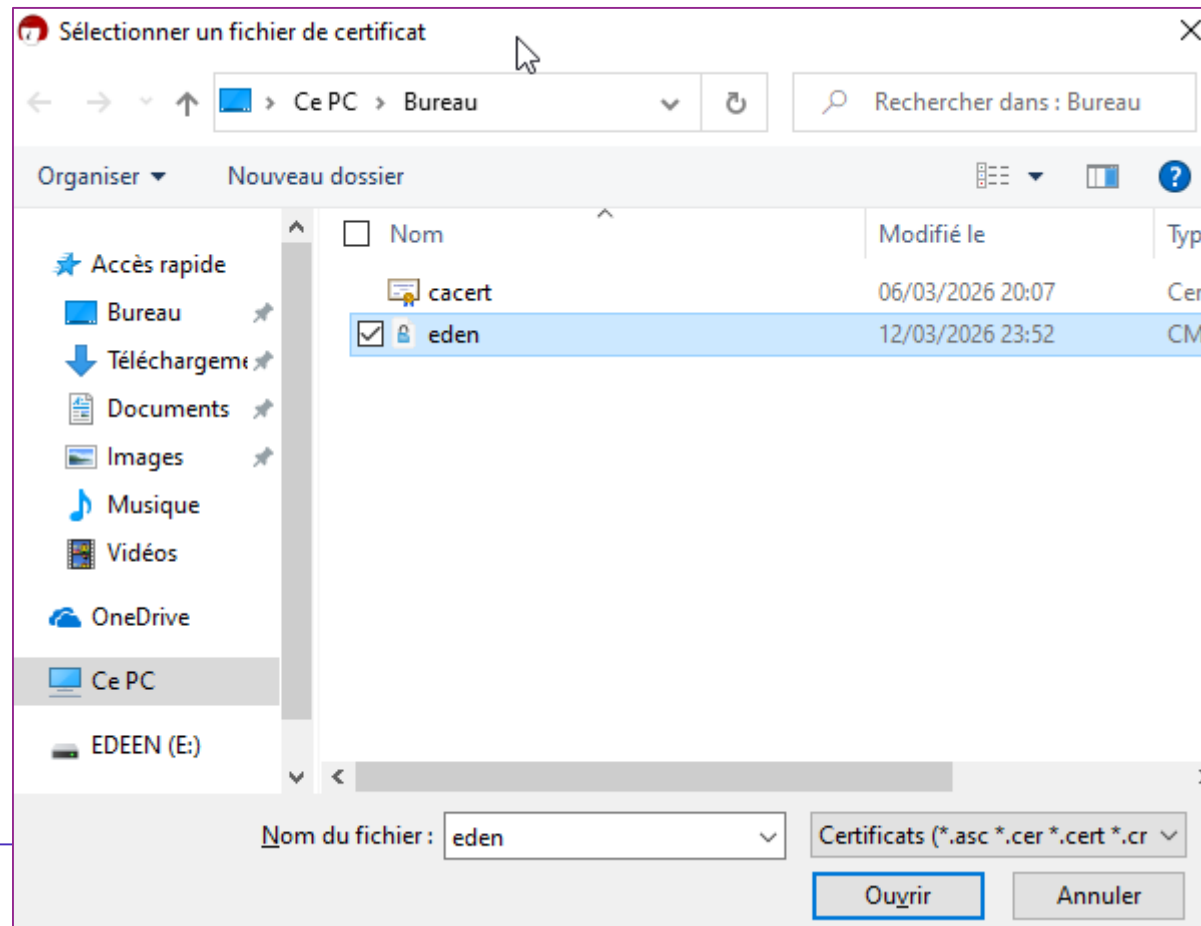
***openssl x509 -outform der -in cacert.pem -out cacert.cer***

## On [Win10\_Client VM], import **cacert.cer** to the **Certificates store**:



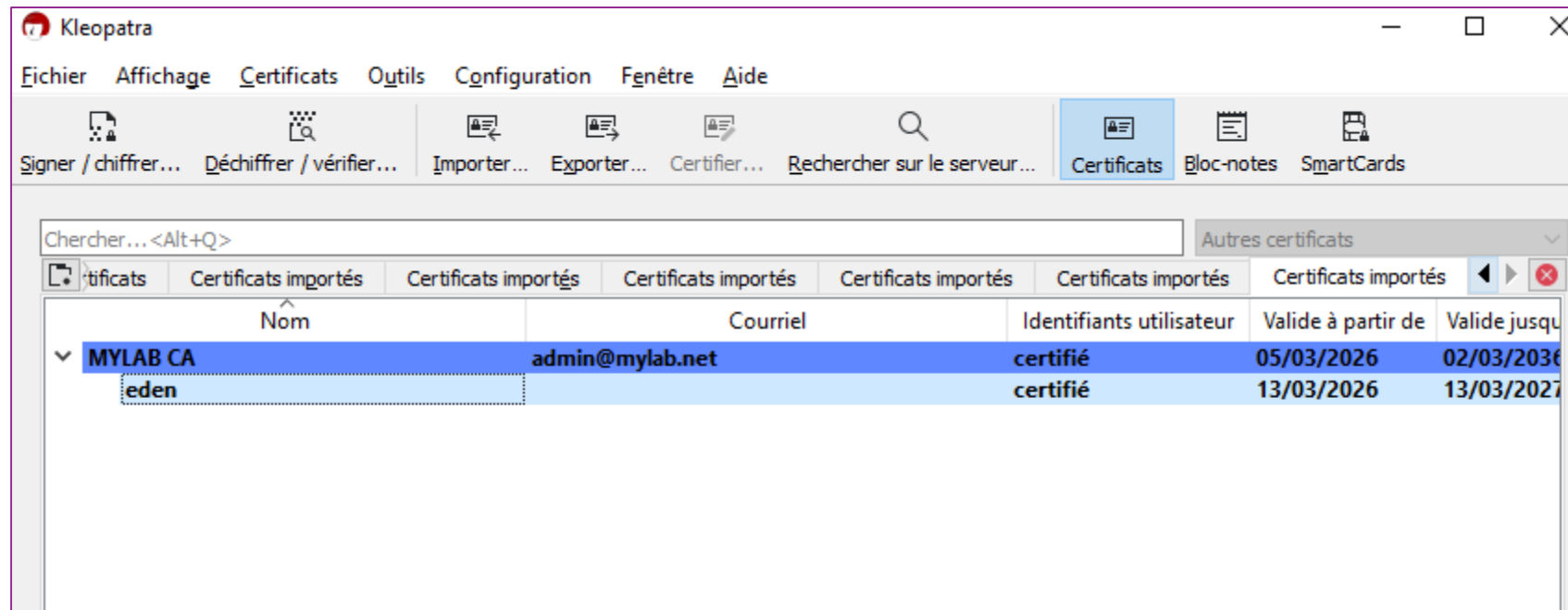
# INSTALL THE USER CERTIFICATE ON KLEOPATRA

## On [Win10\_Client VM], import **eden.pem** to the **Kleopatra** store:



# S/MIME CERTIFICATE INSTALLED

- Once imported, the [eden.pem] certificate [for user eden] is added to Kleopatra, under its CA authority [MYLAB CA] :



# USING S/MIME CERTIFICATE

- Now, user **eden**, can **sign/check** or **encrypt decrypt** files or emails with is **CA-signed certificate** [instead of a **self-signed certificate**].
- If **user1** needs the **public key** of **user2**, he/she just have to request it from the **CA**:
  - By a directory service [**AD** or **OpenLDAP**]
  - By exchanging **signed mail from/to user2**
  - By making **search** on the directory service [**ldap** commands]
- **Note.** No more problems with the **CN/SAN** 😊

## METHOD 2: USING OPENSSL IN WINDOWS

- **For students only** 😊
  - Install **XAMPP** [easy]
  - Install **OpenSSL source** from **Github** [more difficult]